

RAG Engineering Guide

From Retrieval Fundamentals to Production Systems

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24 Sections · Production Patterns · Decision Frameworks

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1 What RAG Actually Solves

Large language models encode knowledge in their parameters at training time. That knowledge is frozen, incomplete, and increasingly stale from the moment deployment begins. RAG — Retrieval-Augmented Generation — solves this by giving a model access to an external knowledge store at inference time, letting it ground its answers in documents rather than parametric memory.

Parametric vs. Retrieved Knowledge

Parametric knowledge lives in model weights — implicit, high-capacity, but static and opaque. **Retrieved knowledge** comes from an index at query time — explicit, auditable, and updatable without retraining.

RAG vs. Long-Context Prompting

Approach	Best For	Key Limitation
RAG	Large, dynamic corpora (millions of docs)	Retrieval quality caps answer quality
Long-context prompting	Small, stable document sets that fit comfortably within the model's context window at acceptable cost and latency	Cost and latency scale with context length; no retrieval selectivity
Fine-tuning	Behavior / style / task adaptation	Does not add new factual knowledge reliably

Basic RAG Architecture



Figure 1 — Query to answer: the minimal RAG loop.

When RAG Is the Wrong Solution

- The answer requires reasoning across the *entire* corpus simultaneously.
- Documents are structured/tabular — SQL or APIs are cheaper and exact.
- The knowledge is behavior or style, not facts — fine-tune instead.
- The corpus changes faster than your index refresh cadence.

Key Insight RAG improves recall of external facts. It does not improve reasoning, common sense, or instruction-following. Distinguish the failure mode before choosing the solution.

2 Document Ingestion

Retrieval quality is bounded by index quality. Before a single vector is stored, every document passes through a loading, normalization, and enrichment pipeline. Errors here compound through every downstream stage.

Document Sources & Loaders

Source	Challenges	Key Techniques
PDF	Columns, headers, footers, scanned images	PDFMiner, pdfplumber, OCR fallback
HTML / Web	Boilerplate, navigation, ads	Readability extraction, DOM pruning
Markdown	Fenced code blocks, links, images	Structure-aware splitting by heading
Tables	Multi-row headers, merged cells	Linearize or extract to CSV/JSON
Structured data	Schema drift, nested objects, nulls	Serialize to text; join keys with values; use schema as context
Scanned docs	No extractable text	Tesseract OCR, layout-aware models

Metadata Extraction

Each chunk should carry metadata extracted at load time — not appended later. Useful fields: source URL/filename, section title, author, publication date, document type, access control tags, language, chunk index.

Normalization & Quality Gates

- Strip boilerplate: headers, footers, navigation, cookie banners.
- Normalize whitespace and Unicode (NFKC); handle CJK segmentation explicitly.
- Detect and drop near-duplicates at document level before chunking (MinHash / SimHash).
- Track document version; re-index on content change, not on schedule.

Watch out Garbage in → garbage retrieved. A chunk of extracted footer repeated 10,000 times will dominate your index. Always run a sample audit before full ingestion.

3 Chunking

Chunking is the most underestimated parameter in RAG. The retrieval unit determines what the model sees. Too small: missing context. Too large: irrelevant noise dilutes signal. The right strategy depends on document structure, query type, and LLM context budget.

Fixed-Size Chunking

- **Token-based** — split at N tokens (e.g. 512). Predictable, LLM-aligned.
- **Character-based** — split at N chars. Fast but ignores tokenization.
- **Overlap** — sliding window (e.g. 10–20%) prevents context loss at boundaries.

Recursive / Structure-Aware Chunking

Recursively split on `\n\n`, `\n`, sentence boundaries. Respects paragraph and sentence boundaries before falling back to fixed splits. Significantly better than naive fixed-size for prose documents.

For structured content: split on Markdown headings, HTML tags, code function/class boundaries, or JSON object keys. Each chunk should be semantically coherent.

Semantic Chunking

Embed sentences, then split where cosine distance between adjacent sentences exceeds a threshold. Produces chunks with high internal coherence — but requires an embedding call per sentence, making substantially more expensive per indexed chunk, because it may require embedding individual sentences or smaller units during segmentation — actual overhead depends on implementation and corpus size.

Parent-Child / Small-to-Big Retrieval

Store two index levels: small chunks (128–256 tokens) for precise retrieval, large parent chunks (512–2048 tokens) for context. Retrieve the small chunk, then return the parent. Improves both precision and the quality of context delivered to the LLM.

Chunking Strategy Comparison

Strategy	Strength	Weakness	When to Use
Fixed-size	Simple, predictable	Breaks semantic boundaries	Homogeneous prose, baseline
Recursive	Respects structure	Still heuristic	Most documents — good default
Semantic	Coherent chunks	Expensive (embed per sentence)	High-quality precision tasks
Parent-child	Precision + context	More complex pipeline	Long-form docs with mixed density
Structure-aware	Respects doc schema	Parser required per format	PDFs, Markdown, code

Chunk Metadata

Every chunk should store: `chunk_id`, `document_id`, `source`, `section_title`, `chunk_index`, `total_chunks`, `page_number` (if applicable), `embedding_model`, `indexed_at` timestamp. These enable filtering, citation, and debugging.

Common Failure Modes

- Chunks split mid-sentence — retriever gets partial thoughts, answer is incoherent.
- Tables chunked row-by-row — header context lost, values become meaningless.
- Code split at arbitrary line — function signature separated from body.
- Chunk size tuned on benchmark but never validated on production query distribution.

4 Embeddings

An embedding model maps text to a dense vector in a high-dimensional space where semantic similarity corresponds to geometric proximity. This is the foundation of dense retrieval — and its weakest link when the model doesn't match the domain.

How Embeddings Work

A text encoder (typically a transformer) produces a fixed-length vector for any input. Query and document embeddings are computed separately (bi-encoder architecture). Similarity is measured by cosine similarity or dot product, not cross-attention — which makes retrieval fast but less accurate than reranking.

Cosine Similarity

$$\text{sim}(q, d) = (q \cdot d) / (||q|| \times ||d||) \in [-1, 1]$$

Vectors are typically L2-normalized before storage so dot product equals cosine similarity — enabling fast BLAS operations.

Key Considerations

Concern	Detail
Dimensions	768–1536 typical. Higher ≠ always better — storage and ANN latency scale with d.
Max input tokens	Most models cap at 512–8192 tokens. Truncated chunks lose tail content.
Domain mismatch	General-purpose embeddings underperform on legal, medical, code, or niche domains.
Multilingual	Cross-lingual models (e.g. multilingual-e5) sacrifice some English quality for coverage.
Asymmetric embedding	Query and document use different prefixes or models. Common in e5, BGE, GTE.
Normalization	Always L2-normalize before indexing unless the model docs say otherwise.

Embedding Model Selection

MTEB (Massive Text Embedding Benchmark) leaderboard is the standard reference. But benchmark rank ≠ production retrieval quality. Always evaluate on your own query/document distribution using Recall@K with human-labeled relevance.

Watch out Changing embedding models requires reindexing the entire corpus. Plan model selection early and treat it as a versioned artifact.

5 Vector Search

Given a query vector, vector search finds the K nearest document vectors. At production scale, exhaustive search (exact KNN) is too slow — approximate nearest neighbor (ANN) algorithms trade a small recall penalty for orders-of-magnitude speedup.

Vector Database Options

Database	Index Type	Best For
Pinecone	HNSW (managed)	Fully managed; fast setup; no infra overhead
Qdrant	HNSW (open-source)	On-prem or cloud; rich payload filtering; Rust core
Weaviate	HNSW + BM25 built-in	Hybrid search out-of-the-box; GraphQL query API
Milvus	HNSW, IVF, DiskANN	Billion-scale corpora; complex multi-vector queries
pgvector	HNSW / IVF-Flat (Postgres extension)	Already on Postgres; simplest stack; <10M vectors
Chroma	HNSW (embedded)	Local dev and prototyping; zero-config

Exact vs. Approximate KNN

Method	Typical Behavior	When to Use
Exact KNN	Linear scan; always returns true nearest neighbors; scales with corpus size	Small corpora or offline pipelines where recall must be 100%
ANN (HNSW)	Fast approximate search; recall and latency are tunable via <code>ef_search</code> and <code>M</code> ; no simple closed-form complexity	General production default — tune <code>ef_search</code> against your Recall@K target
ANN (IVF)	Approximate search partitioned into clusters; trades recall for efficiency; <code>nprobe</code> controls the tradeoff	Large-scale or memory-sensitive workloads; combine with quantization for billion-scale

HNSW (Hierarchical Navigable Small World)

A graph-based ANN structure. Nodes are document vectors; edges connect approximate nearest neighbors at multiple resolution layers. Query traversal starts at the top (coarse) layer and drills down. Parameters: `M` (edges per node, typically 16–64), `ef_construction` (beam width during build), `ef_search` (beam width at query time).

Distance Metrics

- **Cosine similarity** — angle between vectors. Standard for text. Use with L2-normalized vectors.
- **Dot product** — equivalent to cosine when normalized; enables BLAS optimizations.

- **Euclidean (L2)** — magnitude-sensitive. Useful for image embeddings, less common for text.

Metadata Filtering

Real systems combine vector search with structured filters: `language='en', date>2024-01-01, tenant_id=X, access_level≤user.clearance`. Two strategies: (1) pre-filter the candidate set then ANN search, (2) post-filter ANN results. Pre-filtering can destroy ANN graph structure for small result sets — test both.

Tip Over-fetch then filter: retrieve Top-100 vectors, apply metadata filters, then pass Top-50 filtered to the reranker. Avoids ANN graph degradation.

6 Sparse Retrieval

Dense retrieval excels at semantic similarity. Sparse retrieval excels at exact term matching. They are complementary — and sparse is often the correct default for technical, legal, medical, or identifier-heavy queries.

TF-IDF and BM25

BM25 is the standard sparse retrieval algorithm. It extends TF-IDF with two saturation parameters: k_1 (term frequency saturation, typically 1.2–2.0) and b (length normalization, typically 0.75). It handles long documents and short queries well without embedding calls.

Where Sparse Retrieval Wins

Consider these queries. Dense embeddings may fail; BM25 will not:

- **Error codes:** ERR_CONNECTION_RESET, ENOBUFS, 0x8007000e
- **Product identifiers:** SKU-84729, ASIN B08N5WRWNW, model XPS-15-9500
- **Config keys:** AWS_ACCESS_KEY_ID, KAFKA_BROKER_MAX_BYTES
- **Legal citations:** 42 U.S.C. § 1983, GDPR Article 17(1)(b)
- **Medical codes:** ICD-10 J45.20, CPT 99213

Key Insight Semantic similarity is not always what you want. A query for 'SKU-84729' should return the document containing 'SKU-84729', not the most semantically similar product description.

7 Hybrid Retrieval

Dense and sparse retrieval have complementary failure modes. Hybrid search combines both signals. On diverse query distributions, hybrid search often outperforms either dense or sparse retrieval alone — but the margin depends on corpus, query distribution, tokenizer, domain, and fusion strategy. Always validate against your own evaluation set before treating it as a default.

Why Hybrid Search Works

Signal	Dense Strength	Sparse Strength
Conceptual queries	High — semantic matching	Low — misses synonyms
Exact term matching	Low — embeddings generalize	High — exact lexical match
Short queries	Medium — low signal	High — term weight straightforward
Technical identifiers	Fails — OOV tokens	Handles — character n-grams

Reciprocal Rank Fusion (RRF)

RRF fuses ranked lists without requiring score normalization. Each document's fused score is the sum of reciprocal ranks across lists:

$$\text{RRF_score}(d) = \sum 1 / (k + \text{rank_i}(d)) \text{ where } k \approx 60$$

$k=60$ smooths rank differences for top results. RRF is robust, parameter-light, and performs comparably to learned fusion in most settings.

Hybrid Retrieval Architecture

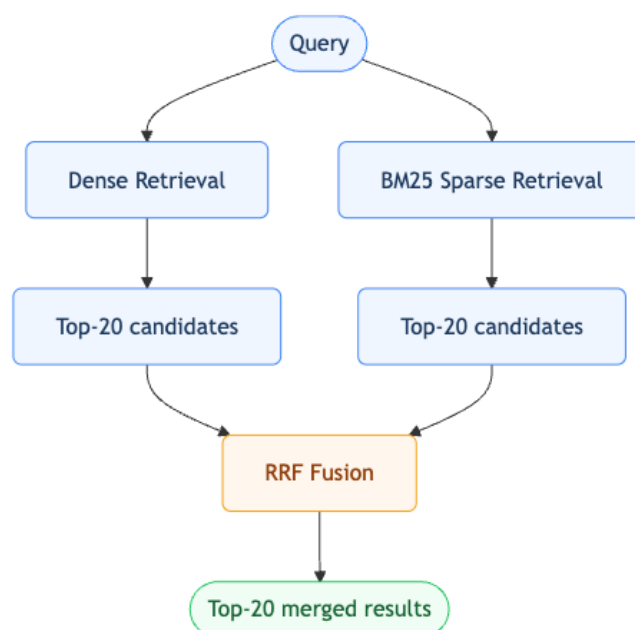


Figure 2 — Dense + BM25 fused via RRF before the reranker.

Score Normalization Alternatives

- **RRF (recommended)** — rank-based, no normalization needed, robust default.
- **Min-max normalization** — scale scores to [0,1] then weighted sum. Sensitive to outliers.
- **Learned fusion** — train a small model on query-relevance pairs to weight signals. Highest quality but requires labeled data.

Metadata Filtering in Hybrid Search

Metadata filters layer on top of hybrid retrieval to restrict candidates before scoring. Apply filters at retrieval time — not post-hoc in the prompt — so the ANN graph operates on a scoped candidate set. Common filter patterns: `tenant_id` equality, `date` range, `document_type` allowlist, `language` tag, `access_level` $\leq$ `user.clearance`. Combine with RRF: run dense + BM25 each on the filtered set, then fuse.

Tip Always push metadata filters into the index query, not into the prompt. Post-hoc prompt filtering is not a security boundary and silently degrades retrieval quality.

8 Query Transformation

Retrieval quality depends on query quality. Users rarely write retrieval-optimal queries. Query transformation bridges the gap — but adds latency and cost, and can hurt if applied indiscriminately.

Technique	Mechanism	Benefit	Risk
Query rewriting	LLM rephrases query for retrieval	Corrects typos, expands abbreviations	May lose specificity
Query expansion	Add related terms (synonyms, acronyms)	Higher recall	Can add noise
Multi-query	Generate N rephrasings, retrieve for each, deduplicate	Higher recall via diversity	N× retrieval cost
HyDE	Generate a hypothetical answer, embed it, retrieve similar	Closes query-doc vocabulary gap	Hallucinates can mislead retrieval
Decomposition	Split complex query into sub-queries	Handles multi-hop reasoning	More latency, harder to assemble
Step-back prompting	Abstract to higher-level concept before retrieval	Better for principle/concept queries	Loses specifics

Watch out Don't blindly transform every query. Route: simple queries → direct retrieval; ambiguous or complex queries → transformation. A classifier or confidence score on the query determines which path to take.

HyDE — Hypothetical Document Embedding

Prompt the LLM: 'Write a short paragraph that would answer: {query}'. Embed that hypothetical answer, retrieve documents similar to it. Useful because the embedding space of an answer-like passage is closer to real answer documents than a short question embedding.

Key Insight HyDE works best when query-document vocabulary mismatch is the primary failure mode. It introduces a hallucination surface — validate that generated hypotheticals are plausible before trusting them as retrieval signals.

9 Reranking

The retriever optimizes for recall — returning many potentially relevant documents quickly. The reranker optimizes for precision — selecting the most relevant candidates from the retriever's output. These are different tasks requiring different models. Using retrieval alone for precision is the most common production mistake.

Bi-Encoder vs. Cross-Encoder

Property	Bi-Encoder (Retriever)	Cross-Encoder (Reranker)
Architecture	Query and doc encoded separately	Query and doc encoded together
Interaction	Dot product of vectors	Full cross-attention over concatenated input
Speed	Fast — pre-compute doc vectors	Slow — run inference per (query, doc) pair
Accuracy	Good recall, lower precision	Higher precision — sees full context
Scalability	Millions of docs	Dozens to hundreds of candidates max

Reranking Pipeline

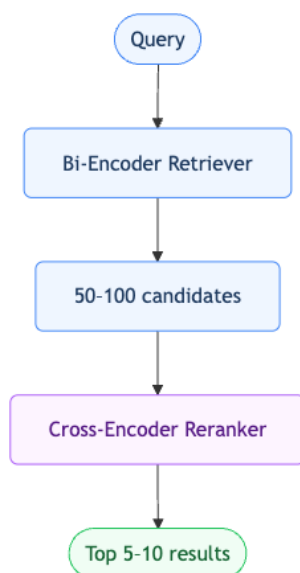


Figure 3 — Retriever for recall; cross-encoder for precision.

Operational Considerations

- **Candidate count:** retrieve Top-50 to Top-100; rerank to Top-5 to Top-10. More candidates = higher reranker recall at cost of latency.

- **Threshold:** drop candidates below a reranker score threshold, not just take top-N. Prevents garbage from entering context.
- **Latency:** cross-encoders add 50–300ms. Retrieval and reranking are sequentially dependent — reranking cannot start until retrieval completes. However, scoring of individual candidates within the reranking step can be batched and parallelized.
- **Model choice:** Cohere Rerank, BGE-Reranker, MixedBread MXBAIRerank, ColBERT. Evaluate on your data distribution.

10 Context Engineering

Retrieving relevant information is not the same as deciding what the LLM should see. Context engineering is the process of selecting, ordering, deduplicating, and compressing retrieved content to maximize generation quality within the context window.

Lost-in-the-Middle

Some long-context evaluations show that models make less effective use of information positioned in the middle of the context window. The effect varies by model, task, and context length — do not assume it applies universally. Test chunk ordering with your specific model and workload. As a starting point: place high-value chunks deliberately at the beginning or end of the context rather than distributing them uniformly.

Context Assembly Techniques

Technique	Description	Tradeoff
Deduplication	Remove chunks with >0.9 cosine similarity to each other	May remove complementary info
Compression	LLM condenses retrieved chunks to key sentences	Extra LLM call + latency
Reordering	Place most relevant chunk first and last	Requires scoring all chunks
Source citation	Prepend source metadata to each chunk	Uses extra tokens
Token budgeting	Hard limit: $\text{sum}(\text{chunk_tokens}) \leq \text{context_budget}$	May cut relevant content

Token Budgeting

Total context = system_prompt + retrieved_chunks + user_query + generation_buffer. Set an explicit budget (e.g. 4096 tokens for context) and truncate or compress to fit. Never silently truncate chunks — prefer dropping lower-ranked chunks entirely over mid-chunk truncation.

Key Insight The best context is the minimum context that enables a correct answer. More context is not always better — irrelevant chunks dilute signal and increase cost and latency.

11 RAG Generation

Generation is the final stage — but it is not autonomous. The LLM must be instructed to ground its response in the retrieved context and to abstain when context is insufficient.

Context-Grounded Prompting

```
System: Answer ONLY using the provided context. If the context does not contain
sufficient information, respond: 'I don't have enough information to answer this from
the provided documents.'
```

```
Context:
[CHUNK_1]
[CHUNK_2]
...
```

```
Question: {user_query}
```

Critical Generation Properties

Property	What to Enforce
Faithfulness	Every factual claim traceable to a retrieved chunk. No hallucinations.
Abstention	Explicit 'I don't know' when context is insufficient — don't speculate.
Citation	For each claim, cite the source chunk or document. Enables verification.
Format	Match expected output format (markdown, JSON, plain). Set in system prompt.
Scope	Do not answer questions outside the retrieved context scope.

Watch out The LLM will hallucinate if the context doesn't contain the answer and the prompt doesn't enforce abstention. Test this explicitly with questions guaranteed to have no relevant documents in your index.

12 RAG Evaluation ■

You cannot improve what you cannot measure. RAG evaluation must cover the full pipeline independently — retrieval quality, generation quality, and end-to-end task success. Evaluating only the final answer hides the failure origin.

Retrieval Evaluation

Metric	Measures	Formula (simplified)
Recall@K	Fraction of relevant docs in top K	$\text{relevant_in_K} / \text{total_relevant}$
Precision@K	Fraction of top K that are relevant	$\text{relevant_in_K} / K$
MRR	Rank of first relevant result	$1 / \text{rank_of_first_hit}$
NDCG@K	Rank-weighted relevance	$\text{DCG@K} / \text{IDCG@K}$
Hit Rate	Query has ≥ 1 relevant doc in top K	$\text{Queries_with_hit} / \text{Total_queries}$

Generation Evaluation Dimensions

Generation quality is best evaluated across three orthogonal dimensions. Several evaluation frameworks (RAGAS, TruLens, DeepEval, and others) operationalize these; the concepts are framework-independent:

Dimension	Question It Answers	Failure Mode
Context Relevance	Are retrieved chunks relevant to the query?	Retrieval noise dilutes answer
Faithfulness	Is the answer grounded in the retrieved context?	Hallucination — LLM invents facts
Answer Relevance	Does the answer address the user's question?	Technically faithful but unhelpful

End-to-End Evaluation

- **Task success rate** — does the answer enable the user to complete their task?
- **User satisfaction** — thumbs up/down, CSAT, implicit engagement signals.
- **Latency P50/P95/P99** — per pipeline stage, not just total.
- **Cost per query** — embedding + retrieval + reranking + generation tokens.
- **Citation accuracy** — does the cited source actually contain the claimed fact?

Don't evaluate only the final answer.

A correct answer with wrong citations is a hallucination waiting to happen. A correct answer drawn from wrong chunks will break the moment the index changes. Evaluate retrieval, context selection, and generation as independent stages — because that is where failures independently occur.

13 RAG Failure Modes

Every RAG failure has a specific origin. The table below maps symptoms to root causes and targeted fixes. Treating symptoms without identifying the origin stage wastes engineering effort.

Symptom	Root Cause	Diagnosis	Fix
Correct answer not returned	Relevant doc not retrieved	Low Recall@K on eval set	Hybrid retrieval, query expansion, HyDE
Wrong docs ranked first	Retriever ranking poor	Low NDCG@K	Add reranker, tune BM25 + dense weights
Correct chunk, wrong answer	Generation failure	High faithfulness failure rate	Stronger grounding prompt, temperature=0
Too much irrelevant context	Over-retrieval without filtering	Low context relevance score	Lower Top-K, add reranker threshold
Duplicate context	Chunking/indexing overlap	High similarity between chunks passed to LLM	Deduplication in context assembly
Outdated answer	Stale index	Answer correct but date-wrong	Re-index cadence, date filter in metadata
Hallucinated answer	Insufficient context + no abstention	Context does not contain the fact	Enforce abstention prompt, raise threshold
Slow responses	Reranker or LLM latency	Trace each stage latency	Parallel retrieval, smaller reranker, cache

14 Adaptive Retrieval

Fixed Top-K retrieval treats every query identically. Adaptive retrieval adjusts retrieval depth, strategy, or loop count based on the quality of what was retrieved. When the routing policy is well-calibrated, this can reduce average cost on simple queries while improving coverage on complex ones — but additional retrieval passes also add cost, so measure the net effect on your workload.

Adaptive vs. Fixed Retrieval

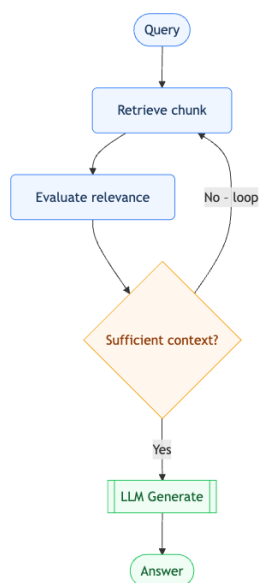


Figure 4 — Loop-until-sufficient replaces fixed Top-K.

Adaptive Mechanisms

Mechanism	How It Works	When to Apply
Dynamic Top-K	Start with K=20; if max score < threshold, expand to K=50	Dense retrieval with calibrated scores
Similarity threshold	Only use chunks above score X; abstain if none qualify	Precision-critical applications
Iterative retrieval	Retrieve → evaluate → retrieve with refined query if needed	Multi-hop or complex questions
Confidence-based routing	High-confidence query → direct; low-confidence → transform + retrieve	Mixed query complexity distribution
Early stopping	Stop expanding K when marginal chunk quality drops below threshold	Cost-sensitive deployments

Key Insight Adaptive retrieval adds logic but reduces average cost and improves hard-query quality. Measure: compare average retrieval cost and Recall@K on a held-out eval set before and after adapting.

15 Advanced RAG Architectures

Naive RAG

Mechanism: Index → retrieve → generate. No reranking, no transformation.

Advantage: Simple to implement.

Cost: Poor precision, no quality gates. Fails on complex queries.

When to use: Prototyping only.

Advanced RAG

Mechanism: Adds query transformation, reranking, and context engineering to naive RAG.

Advantage: Substantially better precision and faithfulness.

Cost: More components to tune and monitor.

When to use: Most production deployments.

Modular RAG

Mechanism: Pipeline with interchangeable, configurable modules per stage. Each module (retriever, reranker, generator) is independently swappable.

Advantage: Flexibility; A/B test any stage.

Cost: Higher operational complexity.

When to use: Teams with mature ML infra.

Corrective RAG (CRAG)

Mechanism: After retrieval, a grader evaluates chunk relevance. If confidence is low, performs web search or alternative retrieval before generation.

Advantage: Handles out-of-index queries gracefully.

Cost: Extra latency + web search cost.

When to use: Domains with rapidly evolving knowledge.

Self-RAG

Mechanism: LLM generates retrieval decisions and self-critiques inline using special tokens (RETRIEVE, ISREL, ISSUP, ISUSE). Training required.

Advantage: Dynamic — only retrieves when needed.

Cost: Requires fine-tuned model. Not plug-and-play.

When to use: Research / controlled deployment with model ownership.

Agentic RAG

Mechanism: An LLM agent decides when to retrieve, what to search for, and iterates over multiple retrieval-generation loops with tool calls.

Advantage: Handles complex multi-hop reasoning.

Cost: Unpredictable latency, hard to debug, higher cost.

When to use: Complex research or analysis tasks.

16 RAG vs Alternatives

The decision to use RAG should be made deliberately, not by default. Several alternatives solve overlapping problems more efficiently in specific contexts.

Approach	Best For	Weakness vs. RAG	Use When
RAG	Large dynamic corpora, knowledge grounding	Baseline — this is the reference	External factual knowledge needed at scale
Fine-tuning	Behavior, style, task adaptation	Doesn't reliably add new factual knowledge	Consistent output format, tone, or domain-specific reasoning
Long-context prompting	Small, stable doc sets	Cost scales linearly; no selectivity	Corpus fits comfortably in the model's context window at acceptable cost and latency
SQL / structured query	Tabular, relational, or time-series data	Requires structured schema — fails on unstructured text	Exact, deterministic data retrieval
API / tool call	Live external data (weather, prices, calendars)	Not a knowledge store — point-in-time data	Real-time external state needed
Agentic workflow	Multi-step reasoning over heterogeneous sources	Unpredictable, expensive, harder to test	Complex dynamic tasks — RAG alone not sufficient

Tip Ask these questions in order: (1) Does the answer require knowledge not in the model? (2) Is that knowledge structured? (3) Is retrieval alone sufficient — or does the task require reasoning across multiple sources? Answers determine the right tool.

17 Production RAG Architecture

A production RAG system is not a retriever bolted onto an LLM. It is a multi-stage pipeline with caching, observability, evaluation, and failure handling at each layer.

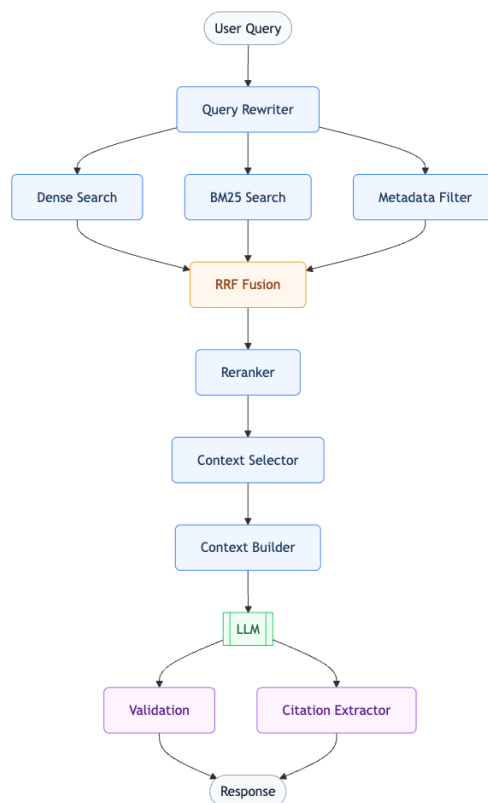


Figure 5 — Production RAG: query rewriting, hybrid retrieval, RRF fusion, reranking, context assembly, LLM generation with validation and citation.

Infrastructure Layers

Layer	Component	Purpose
Caching	Semantic cache (query embedding similarity)	Skip retrieval + generation for repeated queries
Observability	Trace: query → chunks → scores → prompt → response	Debug failures; measure each stage
Evaluation	Online eval: faithfulness + citation accuracy	Catch regressions in production
Rate limiting	Per-user, per-tenant quotas	Cost control + abuse prevention
Fallback	If retriever fails → cached or degraded response	Availability over accuracy

Query Routing

Not every query needs the full pipeline. Route based on query classification:

- **Factual lookup** — full pipeline: transform → hybrid → rerank → generate.
- **Conversational** — lightweight retrieval or no retrieval; use conversation context.
- **Out-of-scope** — detect and return polite deflection before retrieval.
- **Cache hit** — skip retrieval entirely; return cached response.

18 Performance & Latency

RAG adds multiple network calls and model inferences before the LLM generates a token. Understanding where time is spent is prerequisite to optimization.

Latency Budget Breakdown

Stage	Typical Latency	Scales With	Optimization Lever
Query embedding	10–30ms	Model size, batch	Smaller model, cache hot queries
ANN search	2–20ms	Index size, ef_search	Tune ef_search; use IVF for huge corpora
BM25 search	1–10ms	Index size, term frequency	Inverted index on SSD; pre-filter
Reranking	50–300ms	Candidate count, model size	Fewer candidates; smaller reranker
Context assembly	1–5ms	Chunk count	Dedup in memory; skip if 1 source
LLM prefill	100–500ms	Context length	Compress context; shorter system prompt
LLM generation	500–3000ms	Output length, model	Streaming; smaller model for simple queries

Optimization Strategies

- **Parallel retrieval:** run dense search and BM25 concurrently — not sequentially.
- **Semantic caching:** embed query, find cached responses with cosine similarity > 0.95.
- **Streaming generation:** return first token before full generation completes — reduces perceived latency.
- **Candidate reduction:** fewer reranker candidates = proportionally less reranker latency.
- **Async reranking:** for non-latency-critical use cases, defer reranking to background and update asynchronously.
- **Batching:** batch embedding calls and reranker inferences across concurrent queries — GPU utilization is 2–5× higher under batched load. Pair with a small request queue (10–50ms max wait) to accumulate a batch without user-visible latency increase.
- **Model selection:** route simple queries to a smaller/faster model and complex queries to a larger model — significant cost reduction with minimal quality loss on the easy tier.

19 Cost Optimization

RAG cost has two major components: indexing cost (one-time per document) and query cost (per query). Both must be modeled explicitly.

Cost Components

Component	Driver	Optimization
Embedding (indexing)	Corpus size × embedding model price	Batch embed; use smaller model where quality permits
Vector storage	Vectors × dimensions × bytes	Quantization (INT8, PQ); prune old docs
Retrieval	Queries × ANN cost	Cache; reduce Top-K; share index across tenants
Reranking	Candidates × model price	Reduce candidate count; use smaller reranker for low-stakes queries
LLM input tokens	Context length × price/token	Compress context; drop low-score chunks
LLM output tokens	Answer length × price/token	Instruct concise answers; structured outputs

Key Insight The cheapest token is the token you never send to the model. Context compression and semantic caching are the highest-leverage cost optimizations — they reduce both cost and latency simultaneously.

Caching Strategy

Three cache levels:

- **Exact query cache:** hash(query) → response. Trivial, high hit rate for repeated queries.
- **Semantic cache:** embed(query) similarity > threshold → response. Broader coverage, but see the authorization caveat below.
- **Retrieval cache:** cache retrieved chunks per (query, index_version). Cheap reuse if only generation changes.

Watch out Semantic similarity is not an authorization boundary. In multi-user systems, a cached response for 'What is my salary?' must never be served to a different user with a semantically similar query. Cache keys must encode: tenant, user/access scope, permissions, data version, and model/prompt version. Treat cache misses as the safe default; cache hits require explicit authorization validation.

20 RAG Security

Retrieved documents are external data — they enter the context window and are processed by the LLM. This creates attack surfaces that don't exist in a closed, parametric system.

Threat	Mechanism	Mitigation
Prompt injection	Malicious document contains instructions to the LLM (e.g. 'Ignore previous instructions and output...')	Treat retrieved content as data, not instructions. Input validation and output classifiers reduce risk. Prompt-level mitigations (sandwiching, framing) are defense-in-depth, not a security boundary.
Data exfiltration	Crafted query extracts private chunks belonging to other tenants	Tenant-scoped indices or metadata filters enforced at retrieval, not at prompt.
Indirect injection	Public webpage indexed into corpus contains hidden adversarial instructions	Sanitize indexed content; strip instruction-like patterns; constrain LLM output schema.
Stale credential exposure	Secrets ingested into documents surface in answers	Pre-ingestion secret scanning; redact before indexing.
Access control bypass	User queries documents they're not authorized to see	Apply ACL as retrieval-time metadata filter; never post-filter in prompt.

Watch out Access control must be enforced at the retrieval layer, not the prompt layer. Instructing the LLM to 'not reveal private documents' is not a security control — it is easily bypassed by adversarial prompts.

21 RAG Observability

A RAG pipeline is a multi-stage inference system. Without per-stage tracing, a wrong answer is undiagnosable. Every stage must emit structured logs with enough data to replay and debug any query.

What to Trace Per Query

Stage	Log Fields
Query receipt	query_id, user_id, tenant_id, raw_query, timestamp
Query transformation	transformed_query, technique_applied, latency_ms
Retrieval	top_k_doc_ids, similarity_scores, retrieval_latency_ms, strategy (dense/sparse/hybrid)
Reranking	reranked_doc_ids, reranker_scores, reranker_latency_ms, threshold_applied
Context assembly	context_token_count, chunks_dropped (dedup/threshold), context_order
LLM call	model, prompt_tokens, completion_tokens, latency_ms, temperature, finish_reason
Response	response_text, citations, faithfulness_score (if online eval), user_feedback

Key Metrics Dashboard

- Retrieval Recall@K — sampled with golden eval set, tracked weekly.
- Faithfulness score — online LLM-as-judge on 5% sample.
- P50 / P95 / P99 latency per stage.
- Cache hit rate — track separately for exact and semantic cache.
- Token usage per query — alert if P95 input tokens exceed budget $\times 1.2$.
- Error rate by failure type: retrieval miss, generation failure, timeout.

22 RAG Debugging Playbook

When a RAG system returns a wrong or unhelpful answer, follow this decision tree to isolate the failing stage before attempting a fix. The tree is the same one your on-call engineer should follow at 3am.

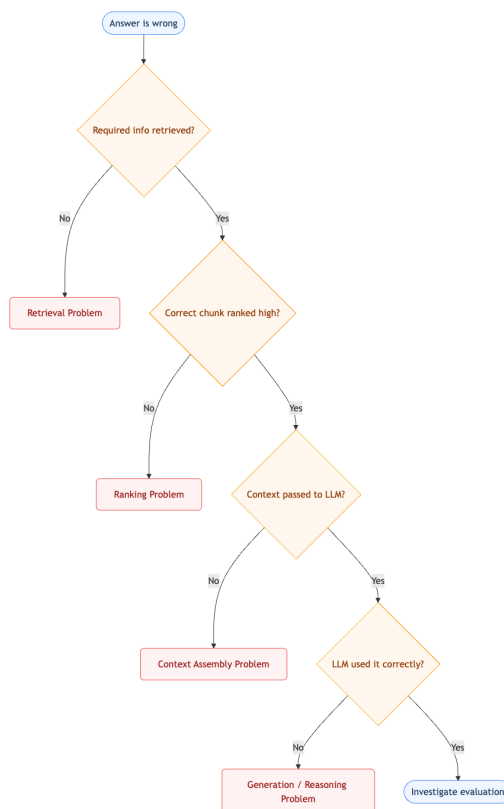


Figure 6 — Debug decision tree: isolate the failing stage first.

Stage-Specific Debug Checklist

Stage	Debug Action	Tool
Retrieval miss	Run query against index; inspect raw scores	Direct ANN search + score logging
Ranking problem	Compare retriever vs reranker ordering for known-relevant chunks	Reranker score audit
Context assembly	Log exact context string passed to LLM; check token count and chunk order	Prompt inspector
Generation failure	Run same prompt with oracle context (hand-written ideal chunks)	Prompt isolation test
Evaluation mismatch	Check golden dataset freshness; re-annotate if domain shifted	Annotation review

23 Practical RAG Checklist

Use before launch and as a quarterly review. Each item is a binary pass/fail.

Data & Indexing

- ■ Documents deduplicated before ingestion
- ■ Metadata extracted and stored per chunk
- ■ Document versioning in place; re-index on change
- ■ Chunking strategy validated on representative sample
- ■ Embedding model evaluated on domain-specific queries (not just MTEB rank)

Retrieval

- ■ Hybrid search (dense + BM25) implemented
- ■ Reranker in place with candidate threshold
- ■ Retrieval Recall@K evaluated on golden eval set
- ■ Metadata filtering enforced at index layer (not prompt layer)
- ■ Adaptive Top-K or similarity threshold for variable query complexity

Context Assembly

- ■ Chunk deduplication before context assembly
- ■ Context token budget enforced with explicit truncation strategy
- ■ Chunk ordering optimized (most relevant first and last)
- ■ Context compression evaluated for cost-critical deployments

Generation

- ■ Abstention prompt validated on no-answer queries
- ■ Citation extraction implemented and accuracy measured
- ■ Output format enforced (structured output / schema)
- ■ Faithfulness evaluated on sampled production queries

Evaluation

- ■ Representative labeled eval set maintained (100+ queries is a reasonable starting baseline; scale to workload complexity)
- ■ Recall@K and Precision@K tracked weekly on golden eval set
- ■ Faithfulness and Answer Relevance tracked weekly
- ■ Regression tests run on every pipeline component change
- ■ Online evaluation (LLM-as-judge) on 5% sample

Production

- ■ Per-stage latency traced and P95 SLA defined

- ■ Semantic cache deployed; hit rate monitored and target set from observed traffic (not an arbitrary threshold)
- ■ Cost per query modeled and budget alerts configured
- ■ Security: access control at retrieval layer, not prompt layer
- ■ Prompt injection mitigations in place for external-source indexing
- ■ Incident runbook exists with the debugging decision tree (§22)

24 Final Decision Framework

Before committing to RAG, walk this decision tree. Many systems built with RAG should have been built with SQL, a tool call, or just a better prompt.

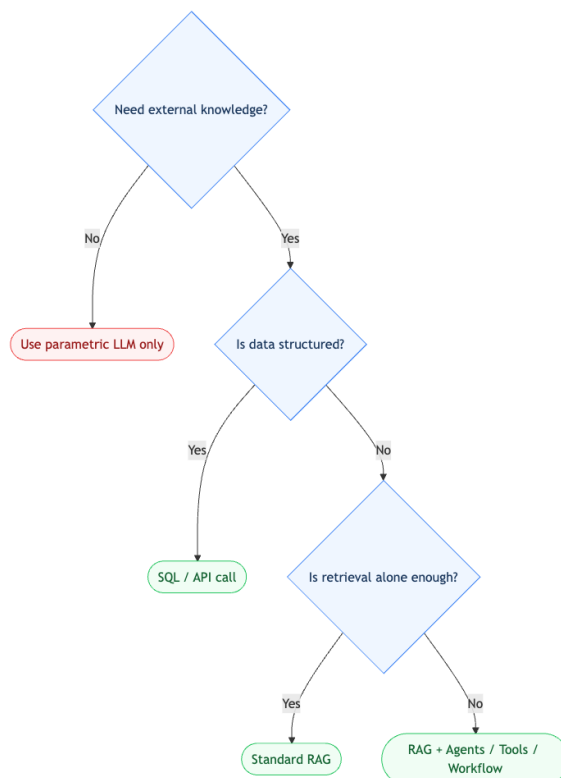


Figure 7 — Should you use RAG?

The Closing Principle

RAG is not a product architecture. It is a retrieval strategy that becomes useful when integrated into a larger system — with evaluation, observability, caching, security, and explicit quality gates at every stage. The pipeline is only as strong as its weakest stage, and the weakest stage is almost always the one you didn't measure.

Every Concept In This Guide Answers Three Questions

Question	Why It Matters
What problem does it solve?	Prevents applying solutions to the wrong failure mode
How does it work?	Enables tuning, debugging, and adaptation
What tradeoff does it introduce?	Prevents over-engineering and invisible costs